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HILOM: Hospital Inflow and Optimization Model

Trace the Pattern,
Target the Cure:
Advancing Public Health
Through Data Science

503: Sleep Unavailable

Kathlyn Mae Postrado

Neil Ivan Cajipe

Alexia Nicole Lynne Dalusong

Reese Chloe Santiago

Justin Kurt Tin

EXECUTIVE SUMMARY

At one Philippine public hospital's outpatient department, patients averaged 4 hours and 41 minutes per visit, nearly four hours of it waiting and 42 minutes in consultation (Andres et al., 2023). Such delays lead some patients to abandon care and others to worsen while they wait (Alejo, 2021). Weak primary care pushes patients to bypass local facilities and crowd emergency departments (Barbadillo et al., 2024), yet existing referral and navigation systems still coordinate diversions manually and only after emergency units are full. This project proposes a system that anticipates surges and diverts non-critical patients before emergency rooms reach capacity. The system will forecast hourly patient arrivals 24 hours to 7 days ahead using SARIMA/SARIMAX and gradient-boosted regression, targeting MAPE and WAPE below 15%, MAE ≤ 2 patients/hour, and RMSE ≤ 3 patients/hour against a seasonal-naive baseline. Queuing models (M/M/c, with M/G/c or discrete-event simulation as alternatives) will estimate waiting times and identify potential capacity breaches at least 24 hours in advance, targeting waiting-time MAE ≤ 10 minutes and RMSE ≤ 15 minutes, while a constrained optimization model will recommend diversion volumes and destinations for non-critical patients to keep projected waiting times below 45 minutes, targeting at least a 10% reduction in mean waiting time and a 15% reduction in capacity-violation rate relative to baseline operations. Administrators and local government officials will review the results on a web dashboard, test diversion scenarios, and receive a protocol memo automatically drafted by a large language model (LLM) for human review. Retrospective testing and simulated stress scenarios will compare current operations against optimized diversion. As a decision-support tool, the system excludes high-acuity resuscitation cases, may not anticipate mass-casualty events or other anomalous surges, and does not override clinical judgement or hospital protocols.

KEYWORDS

Patient Inflow Forecasting, Queuing Theory, Hospital Wait Times, Emergency Department Overcrowding, Clinical Load Balancing, Resource Allocation, Automated Triage Policy

BACKGROUND

In major Philippine hospitals and referral centers like the Philippine General Hospital (PGH), patients may experience delays in accessing basic medical attention. In an assessment of a public hospital outpatient department, Andres et al. (2023) observed that patients spent an average of 4 hours and 41 minutes inside the facility, with nearly 4 hours (238.58 minutes), about 85% of the visit, spent waiting in queues, compared with 42.11 minutes of actual consultation. The study associated these persistent delays with reduced patient satisfaction across all service dimensions, largely driven by intake bottlenecks at the triage desk. Furthermore, these prolonged waiting times are systemic across the Philippine health system, where severe congestion routinely compels exhausted patients to abandon appointments without receiving care (Alejo, 2021).

Compounding this issue, Barbadillo et al. (2024) emphasized that weaknesses in the broader healthcare network force patients to bypass local facilities, further overburdening hospital emergency departments. Their study also cited prior PGH and tertiary hospital time-motion studies where turnaround times ranged from 109 and 117 minutes. Nevertheless, operational interventions may help improve patient flow. For instance, Andres et al. (2023) demonstrated that balancing patient triage reduced waiting times by 46% and increased intake capacity by 86%. Similarly, Alejo (2021) and Barbadillo et al. (2024) underscored how eliminating process waste and tracking patient flow using digital records can help sustain manageable turnaround times. Despite these operational insights, existing referral and navigation systems still rely on hotline-based coordination and centralized command centers (Department of Health, 2022). Because transfers are coordinated based on current capacity reports rather than predictive demand forecasting, redirection occurs reactively only after facilities reach capacities.

THE PROBLEM

Excessive wait times can place additional pressure on patients and healthcare facilities, potentially turning operational inefficiencies into active health risks as treatable illnesses worsen and patients leave without receiving care and treatment (Alejo, 2021; Andres et al., 2023). These delays are compounded by intake bottlenecks, reactive and poorly coordinated referral processes, and limited predictive capacity to anticipate surges in patient arrivals (Alejo, 2021; Andres et al., 2023; Barbadillo et al., 2024). Developing a digital system capable of balancing the load in the current hospital and healthcare facilities is therefore essential to redirect non-critical patients to the correct facilities and optimize overall healthcare distribution.

OBJECTIVES

To address the systemic problem of hospital congestion, this project aims to design, develop, and validate an automated clinical load-balancing system that combines time series arrival forecasting with multi-server queuing theory to predict emergency department wait times and recommend the timely diversion of non-critical patients to appropriate healthcare facilities based on projected patient demand and service capacity.

Develop an Inflow Forecasting Model: Train and evaluate seasonal-naive, Seasonal Autoregressive Integrated Moving Average (SARIMA)/SARIMA with Exogenous Variables (SARIMAX) and gradient-boosted regression models using historical patient arrivals over rolling horizons of 24 hours to 7 days ahead, targeting Mean Absolute Percentage Error (MAPE) and Weighted Absolute Percentage Error (WAPE) below 15%, Mean Average Error (MAE) ≤2 patients/hour, and Root Mean Square Error (RMSE) ≤3 patients/hour, with model performance compared against the seasonal-naive baseline using chronological train-validation-test splits and rolling-origin evaluation.

Model Dynamic Waiting Times: Formulate an M/M/c queuing model, extending to M/G/c or discrete-event simulation (DES) arrival rates, historical service durations, and available staffing levels to estimate waiting times, quantify server utilization, and flag projected exceeding of capacity threshold at least 24 hours prior to department saturation. where its assumptions do not hold.

Build Triage Optimization Engine: Implement a constrained optimization algorithm ("triage inverter") to calculate optimal diversion percentages and

destinations for non-critical cases, keeping wait times under 45 minutes and nurse-to-patient ratios within institutional standards.

Automate Policy Memos: Use an LLM API to translate optimization outputs and user-defined constraints into structured, actionable triage protocol memoranda for hospital administrators and LGU officials within 10 seconds of execution.

Build an Interactive Dashboard: Build a web-based interface that integrates the backend models, allowing health administrators to simulate diversion scenarios and visualize patient load within a few seconds under defined test conditions.

SCOPE & LIMITATIONS

This project will develop predictive models and an optimization pipeline to support the load balancing of non-critical patients across internal clinics and local health networks over 24 hours to 7 days. The routing system excludes high-acuity resuscitation cases. Furthermore, the system relies on historical intake data, which may not reliably anticipate unprecedented mass-casualty events or other anomalous surges, and functions strictly as a decision-support tool that does not override clinical judgment or supersede established hospital protocols and statutory requirements.

METHODOLOGY

This study will employ a quantitative, computational research design integrating time-series forecasting, queuing theory, constrained optimization, and discrete-event simulation (DES) to develop and evaluate a predictive load-balancing system. The system will forecast patient arrivals, estimate congestion, and recommend diversions of non-critical patients across participating healthcare facilities.

Data Acquisition and Preprocessing

Historical patient records will be collected subject to institutional approval and privacy requirements, including arrival timestamps, service durations, patient acuity, staffing levels (including nurse-to-patient ratios), facility capacity, and referral records. Data will be standardized and aggregated into hourly intervals, with separate waiting-time measures for triage and consultation where available.

Patient-Arrival Forecasting

Seasonal-naive, SARIMA/ SARIMAX, and gradient-boosted regression models will be evaluated to forecast hourly arrivals over rolling 24-hour to 7-day horizons. Chronological train-validation-test splits and rolling-origin evaluation will be used to prevent temporal data leakage.

Dynamic Queuing and Capacity Estimation

An $M/M/c$ queuing model will estimate waiting times, queue lengths, and server utilization using forecasted arrivals, service durations, and available staffing. An $M/G/c$ model or DES will be considered when service-time distributions or time-varying patient flows violate simplifying assumptions. DES has been used to redesign hospital service systems and reduce emergency department waiting times (Mandahawi et al., 2010; Siska & Morena, 2014), and to test process changes before implementation (Arafeh et al., 2018). Predictions will be validated against observed waiting times to establish operational thresholds and identify potential capacity breaches.

Constrained Diversion Optimization

A constrained optimization model will recommend diversion volumes and

destination facilities by minimizing projected waiting-time penalties, travel times, and receiving-facility congestion. Constraints will account for demand, clinical appropriateness, service availability, travel requirements, staffing, and receiving capacity. High-acuity resuscitation cases will be excluded, and recommendations will remain subject to authorized clinical review.

System Integration and Dashboard Development

The forecasting, queuing, and optimization models will be integrated into a web-based dashboard displaying projected arrivals, waiting times, capacity alerts, and baseline-versus-optimized scenarios. An LLM API will generate draft policy memoranda from verified model outputs and approved institutional protocols, targeting generation within 10 seconds and requiring human review before implementation.

System Validation and Evaluation

The system will be evaluated through retrospective testing and DES, comparing baseline operations with optimized diversion under equivalent demand and staffing conditions. Forecasting accuracy will be assessed using MAE (≤2 patients/hour), RMSE (≤3 patients/hour), and WAPE and MAPE (<15%). Waiting-time estimates will be evaluated against observed values using MAE (≤ 10minutes) and RMSE (≤15 minutes). Early-warning performance will be assessed using advance warning time, breach detection rate, and false-alarm rate, targeting at least 24 hours of advance notice for anticipated capacity breaches, detection of at least 90% of breaches before occurrence, and a false-alarm rate ≤20% of issued alerts. Operational effectiveness will be measured through mean waiting time and capacity-violation rate, targeting reductions of ≤10% and ≤15%, respectively, while projected waiting times will be kept under 45 minutes. Forecast generation and alert-generation times will be limited to 60 seconds. All diversion recommendations will remain subject to capacity, staffing, clinical-routing constraints, institutional protocols, and authorized clinical review.

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